BT 403 GDB Spring 2021

What is plant stress? How will you differentiate the biotic and abiotic stress? Also discuss the effect of stress on productivity losses?

Plant stress can define as the condition in the environment which included external conditions that adversely affect growth, development and productivity of plants.

Biotic stress included the living factor which causes the affect the growth of plant like fungi and virus. Biotic stress is caused by living things like microorganisms included pathogens release of allelochemicals by the plant and another organism

Abiotic or non-biotic stress are adverse or negative effects of the physical and chemical components of the environment arising from the interaction with living organisms. non biotic are cause by changes in environmental factors like lack of soil nutrients, shortage of water, acidic soils and high salts.

Effects of stress on Productivity:

Stress can cause the loss of productivity, because it can damage the plant by changing the condition which is not suitable for the plant and they cannot growth such a kind of condition. Crop yield is mainly influenced by climatic factors, agronomic factors, pests and nutrient availability in the soil. Stress is any adverse environmental condition that hampers proper growth of plant. Crop productivity is significantly affected by drought, salinity and cold. Abiotic stress reduces water availability to plant roots by increasing water soluble salts in soil and plants suffer from increased osmotic pressure outside the root. Physiological changes include lowering of leaf osmotic potential, water potential and relative water content and creation of nutritional imbalance, enhancing relative stress injury or one or more combination of these factors.

VU Medical Zone effort by Amaan Khan

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